

Algebra II with Trigonometry

Chapter 5.4

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

Domain of Inverse Cosine What is the domain of the inverse cosine function, $\cos^{-1}(x)$?

Range of Inverse Tangent What is the range of the inverse tangent function, $\tan^{-1}(x)$?

Evaluating Inverse Sine What is the exact value of $\sin^{-1}(0)$ in radians?

Evaluating Inverse Cosine What is the exact value of $\cos^{-1}(-1)$ in radians?

Evaluating Inverse Tangent What is the exact value of $\tan^{-1}(-1)$ in radians?

Reciprocal Inverse Identities How can you rewrite the expression $\sec^{-1}(5)$ using the inverse cosine function?

Evaluating Validity Based on its domain constraints, is the expression $\sin^{-1}(-2)$ defined?

Mental Right Triangles Using your knowledge of 3-4-5 right triangles, what is the exact value of $\cos(\sin^{-1}(4/5))$?

Mental Trig Composition Using a mental right triangle, what is the exact value of $\cos(\tan^{-1}(4/3))$?

Core Concept of Inverses Why must the standard sine function's domain be restricted to $[-\pi/2, \pi/2]$ in order to define an inverse function?